

Physics-aware neural surrogates for high-fidelity ultra-fast simulation of nanophotonic sensors

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Abstract

Nanophotonic sensors rely on spatially localized electromagnetic fields for signal transduction, but full-wave simulation of these fields is computationally expensive for large-scale design exploration. Existing machine-learning methods for metasurfaces mainly predict scalar optical responses, leaving spatial field prediction for fabrication-aware nanophotonic sensors largely unexplored. Here we introduce MetaField, a physics-aware neural surrogate that directly learns geometry-to-field mappings from periodic MetaSPR structures. The framework combines a fabrication-aware simulation database, physics-aware tensor representations and a boundary-consistent neural architecture to predict dominant full-wave field distributions under TM-like illumination across variable sizes and dispersive materials. Over the visible spectrum, MetaField reproduces full-wave fields with high spatial consistency while achieving

orders-of-magnitude faster inference than conventional FDTD on diverse hardware. Integrated with a surrogate-assisted genetic algorithm, MetaField enables inverse design evaluating $>10^6$ candidates in hours instead of years of brute-force optimization. Experimental validation with nanopillar MetaSPR structures shows that MetaField-predicted near-field enhancement trends match measured SERS responses. Thus, MetaField provides a field-resolved neural surrogate for ultra-fast nanophotonic sensor simulation, inverse design and field-guided analysis.

Keyword: MetaSPR; nanophotonic sensors; electromagnetic field prediction; neural surrogate

Introduction

Nanophotonic sensors rely on subwavelength electromagnetic confinement¹ to convert local material or molecular perturbations into measurable optical responses, including resonance shifts², scattering signatures³ and surface-enhanced Raman spectra⁴. Although such responses are often reported as spectra, their physical origin lies in spatially localized near-field distributions shaped by geometry, material dispersion and boundary conditions^{5–7}. For plasmonic and metasurface-based sensors⁸, quantities such as hotspot location⁹, field intensity within analyte-accessible regions¹⁰ and modal overlap near metal–dielectric interfaces are central to sensing performance¹¹. Accurate field-resolved modelling is therefore a computational prerequisite for interpreting sensor responses and for exploring nanophotonic design spaces beyond empirical trial-and-error^{12,13}. Full-wave electromagnetic solvers, such as finite-difference time-domain and finite-element methods, remain the standard tools for modelling nanophotonic sensors because they resolve dispersive materials, multilayer interfaces and spatial electromagnetic fields without reducing the response to a few scalar optical observables^{14,15}. In practice, even geometry-resolved nanometer-scale simulations become computationally expensive when repeated across many wavelengths, material stacks and structural variants¹⁶. This burden is amplified when the desired output is not only a reflection or transmission spectrum, but the full spatial

distribution of the near field for each candidate design¹⁷. Such field-resolved simulations are essential for hotspot optimization, inverse design and sensing-region analysis, yet they are difficult to use as the inner loop of high-throughput design exploration¹⁸. Full-wave simulation therefore remains indispensable as a ground-truth verifier, but its cost motivates fast surrogate models that can predict spatial field distributions directly¹⁹.

Recent advances in machine learning have substantially accelerated nanophotonic modelling and inverse design by learning mappings between structural parameters and optical responses^{19,20}. Most metasurface-oriented predictors, however, are formulated to output transmission²¹, reflection²², absorption²³, phase profiles²⁴ or scattering spectra^{25,26}, which are effective for rapid performance evaluation but leave the spatial electromagnetic fields implicit. As a result, these models provide limited direct access to sensing-critical quantities such as hotspot morphology, analyte-region field enhancement and mode localization. Consequently, Inverse-design networks built on such response predictors similarly tend to optimize macroscopic optical targets rather than field distributions within physically relevant regions^{27,28}. To address this limitation, neural surrogate solvers, neural operators and physics-informed models have begun to predict electromagnetic field distributions directly^{29–31}. Representative examples include near-/far-field predictors for arbitrary nanostructures and physics-augmented surrogate solvers for photonic devices^{32–34}. However, these approaches are typically developed for general scattering systems, dielectric photonic devices or free-form geometries, and do not explicitly address fabrication-aware periodic nanoplasmonic sensor structures with multilayer stacks, mixed electromagnetic boundary conditions, variable simulation sizes and source-dependent phase conditioning^{35–37}.

Modern deep-learning architectures provide useful building blocks for electromagnetic field prediction, including encoder–decoder networks, attention mechanisms and conditional modulation^{38–40}. Yet architecture alone is insufficient for nanophotonic sensor simulation: field prediction also requires fabrication-aware data

representations, broadband material dispersion, boundary-consistent learning and validation against both full-wave simulation and experimental sensing readouts^{41–43}. This creates a need for a field-resolved, physics-aware surrogate framework that combines neural-network expressivity with sensor-specific electromagnetic structure, data construction and verification.

Here we introduce MetaField, a physics-aware neural surrogate for high-fidelity, ultra-fast simulation of nanophotonic sensors (Figure 1). Rather than predicting only global optical responses, MetaField learns spatially resolved full-wave electromagnetic field distributions from fabrication-aware MetaSPR sensor structures. The framework combines fabrication-aware representations and a neural geometry-to-field mapping for accelerated field prediction. Full-wave simulation is retained as the verification standard, while MetaField provides the inference speed required for large-scale design-space screening and field-based sensor analysis. This study establishes MetaField through three linked components: MetaSPR-SimDB, a fabrication-aware full-wave simulation database and encoding strategy. MetaField, a physics-aware neural architecture combining multiscale convolution, conditional modulation and attention-based long-range modelling and a validation workflow connecting accelerated field prediction to FDTD-verified inverse design and SERS-oriented experimental trend analysis. Together, these components shift metasurface machine learning from optical-response prediction toward spatially resolved electromagnetic-field learning. MetaField thereby provides a computational basis for ultra-fast, field-guided nanophotonic sensor simulation and design.

Results and Discussion

MetaField framework

The MetaField framework consists of three tightly coupled components: the MetaSPR-SimDB simulation database, the neural surrogate, and downstream field-guided applications (Fig. 1).

MetaSPR-SimDB (Fig. 1a) was constructed from full-wave FDTD simulations of

periodic MetaSPR nanophotonic sensor structures under TM-like illumination. Under the present two-dimensional x-z simulation setting, the dominant field components are E_x , E_z and H_y , whereas E_y , H_x and H_z remain negligible (Fig S1-S2).

The MetaField architecture (Fig. 1b) establishes a geometry-to-field mapping from physics-aware input tensors to full-wave electromagnetic field distributions. Input representations combine complex permittivity channels, spatial phase coordinates and geometry-aware distance fields together with global physical conditions. The network adopts a multiscale encoder–decoder structure with mixed-boundary convolutions, FiLM-based conditional modulation and a transformer bottleneck module. The network

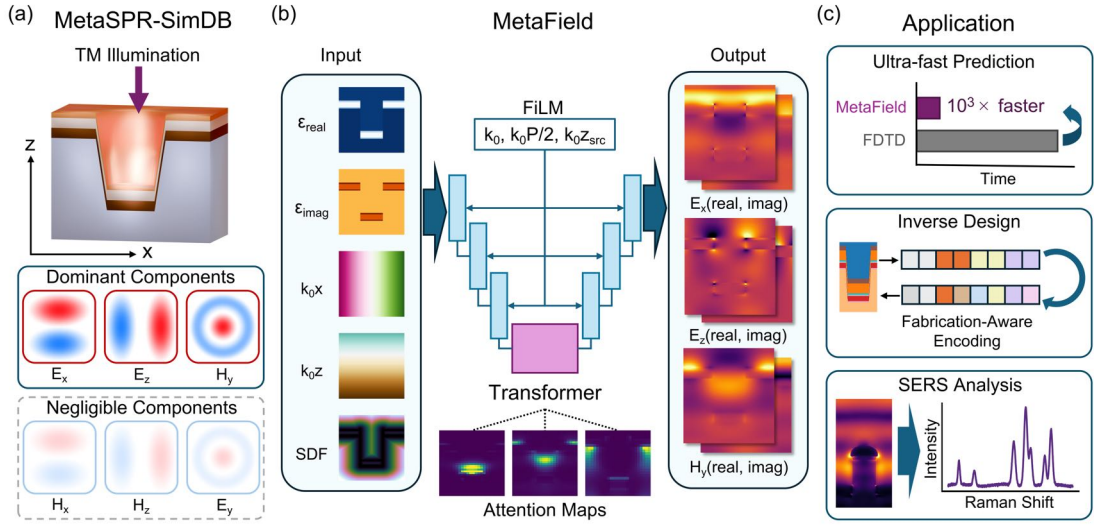


Figure 1 Overview of the MetaField framework. (a) MetaSPR-SimDB provides fabrication-aware periodic nanophotonic sensor structures under TM-like illumination together with dominant full-wave electromagnetic field components. (b) MetaField maps physics-aware inputs and fabrication-aware structure representations to spatial electromagnetic field distributions through a neural surrogate framework with conditional modulation and long-range modelling. (c) The trained surrogate enables ultra-fast field prediction, inverse design and SERS-oriented analysis.

combines mixed-boundary convolution and transformer-based modelling within an encoder–decoder framework.

The trained surrogate further enables several field-guided applications (Fig. 1c). First, MetaField provides orders-of-magnitude faster field prediction than conventional full-wave simulation. Second, the surrogate can be integrated with fabrication-aware inverse-design workflows to support large-scale design-space exploration. Third, predicted near-field distributions can be connected to experimentally measured sensing

responses, enabling SERS-oriented electromagnetic field analysis.

MetaSPR-SimDB and fabrication-aware encoding

To support field-resolved learning for periodic nanophotonic sensors, we constructed MetaSPR-SimDB, a fabrication-aware full-wave electromagnetic simulation database containing 70,000 samples from 175 MetaSPR structures over 400 wavelengths. The database was designed to preserve both electromagnetic field information and fabrication-related structural representations. Figure 2a illustrates the

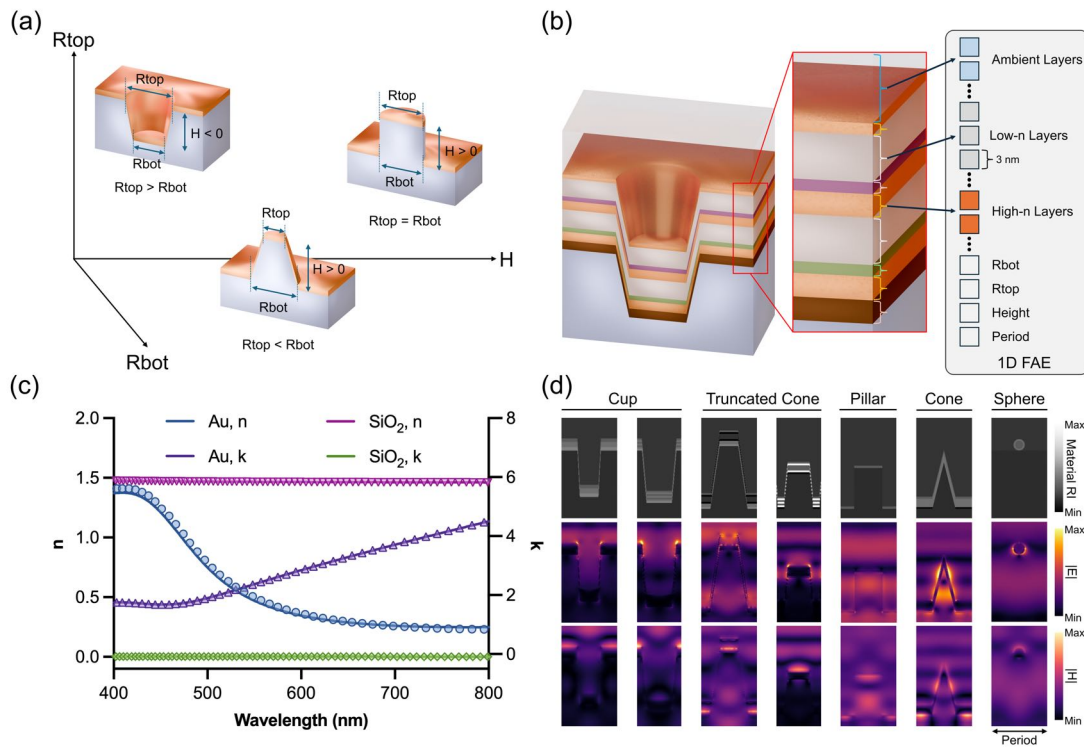


Figure 2 MetaSPR-SimDB and fabrication-aware encoding. (a) Geometry parameterization of periodic MetaSPR structures using R_{top} , R_{bot} and height H . (b) Fabrication-aware encoding (FAE) of multilayer MetaSPR structures and physical conditions. (c) Representative dispersive material modelling using wavelength-dependent refractive-index fitting. (d) Representative periodic nanophotonic structures together with corresponding electromagnetic field distributions.

geometric parameterization used for periodic MetaSPR structures. The structural space includes nanocup-, nanopillar- and tapered-type geometries represented through top radius R_{top} , bottom radius R_{bot} , height H and structural period P . Positive and negative values of H correspond to protrusive and cavity-type structures, respectively.

To represent fabrication-constrained multilayer structures in a unified format, a

fabrication-aware encoding (FAE) strategy was introduced (Fig. 2b). The encoding combines multilayer material slots and geometries within a compact tensor representation. Material dispersion was incorporated through wavelength-dependent refractive-index modelling. Dispersive optical properties were parameterized using Drude–Lorentz representations fitted to experimental refractive-index spectra, with up to 12 Lorentz oscillators used in the current implementation. Figure 2c shows representative broadband dispersion curves used in the database.

MetaSPR-SimDB contains a broad range of periodic nanophotonic structures, including nanocups, truncated cones, nanopillars and related plasmonic geometries (Fig. 2d). For each structure, full-wave electromagnetic field distributions were recorded under TM-like illumination rather than only scalar optical responses. This formulation establishes MetaSPR-SimDB as a field-oriented nanophotonic learning dataset for surrogate modelling and inverse design.

Physics-aware MetaField architecture

The MetaField framework was developed as a physics-aware neural surrogate for spatial electromagnetic field prediction in periodic nanophotonic sensor structures. Figure 3 shows the MetaField architecture based on an encoder–decoder framework with mixed-boundary convolution, FiLM conditioning and a transformer bottleneck.

Physics-aware input tensors consisted of complex permittivity distributions, spatial phase coordinates and geometry-aware normalized distance fields. Spatial phase coordinates k_0x and k_0z were included in the input representation. An X-centered coordinate representation was further adopted to maintain periodic symmetry around the structural center. To support periodic nanophotonic structures with different spatial sizes, MetaField employed multi-size grouped training together with mixed-boundary convolutions. Horizontal boundaries used circular extension along the periodic direction, while vertical boundaries used replicate extension along the propagation direction. This mixed-boundary treatment was implemented through the PhysicsConv2d operator. The network backbone adopted a multiscale encoder–decoder

structure with FiLM-conditioned residual blocks. Global physical conditions, including normalized wavenumber, structural period and source-position phase terms, were injected through conditional residual modulation. A transformer bottleneck module with relative-position-aware attention was inserted between encoder and decoder stages.

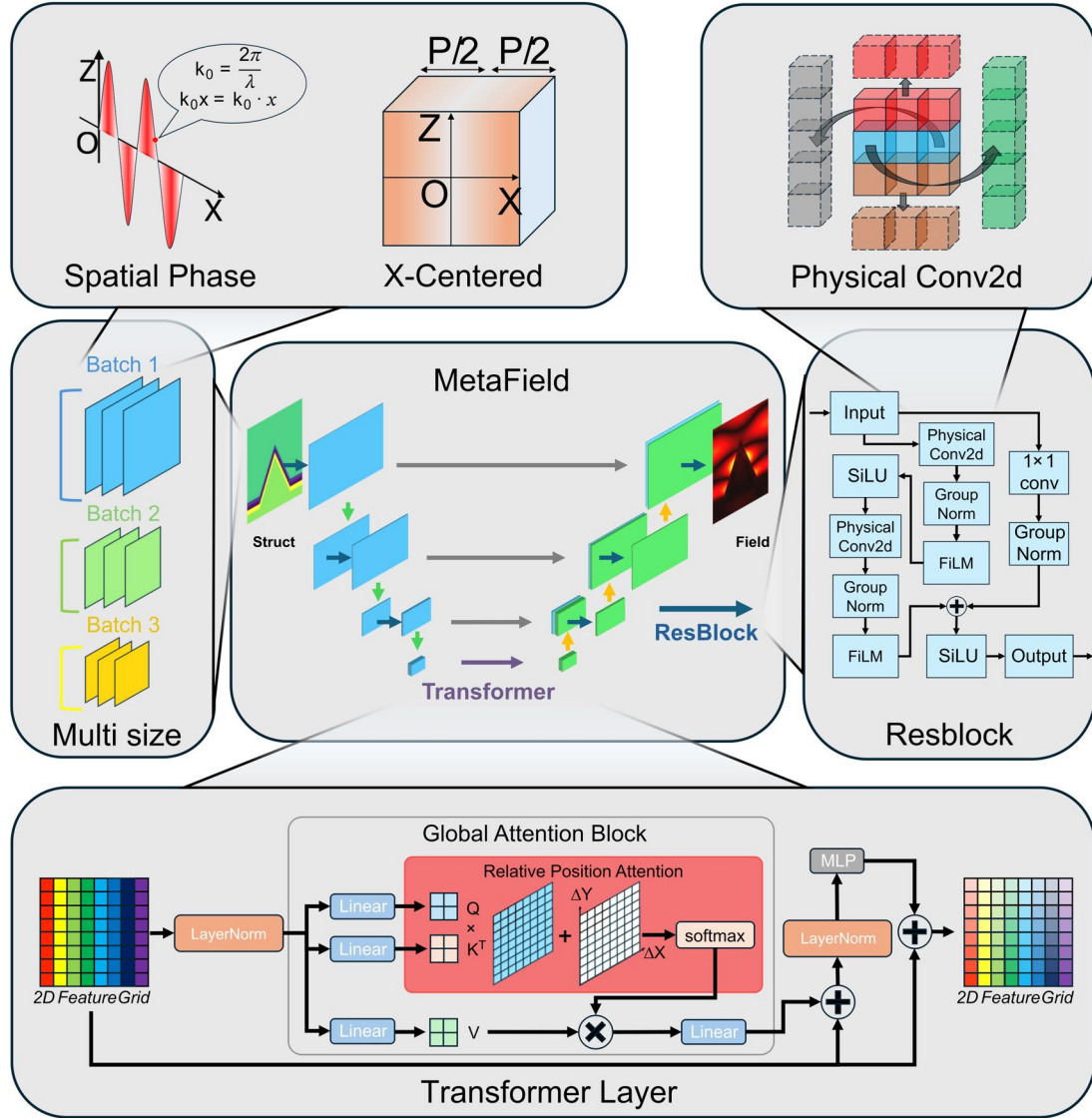


Figure 3 Physics-aware MetaField architecture. MetaField combines physics-aware spatial representations, mixed-boundary convolution, conditional residual modulation and transformer-based long-range modelling within a multiscale encoder–decoder framework. The network incorporates spatial phase coordinates, X-centered representations and multi-size training together with PhysicsConv2d, FiLM-conditioned residual blocks and relative-position-aware attention.

High-fidelity field prediction and acceleration

The predictive performance of MetaField was evaluated using spatial

electromagnetic field reconstruction across the validation dataset. Figure 4a shows the wavelength-dependent prediction error for the dominant field components $|E|$ and $|H|$. Across most of the visible spectral range, MetaField maintained stable reconstruction accuracy with low mean MSE values and limited variance. Increased prediction error was observed near strongly resonant spectral regions, where localized field enhancement and rapid spatial field variation become more pronounced. Inference throughput was benchmarked against conventional full-wave simulation workflows using multiple hardware configurations (Fig. 4b). Compared with CPU- and GPU-based FDTD simulations, MetaField achieved orders-of-magnitude faster inference while directly predicting spatial electromagnetic field distributions. The surrogate maintained efficient inference performance across consumer GPU (NVIDIA RTX 3060), server GPU (NVIDIA A100 80GB) and Apple Silicon (M5 Max) hardware platforms.

Representative field comparisons are shown in Figure 4c. MetaField accurately reproduced the electromagnetic fields, including hotspot localization, boundary-enhanced field confinement and near-field coupling distributions. Residual field maps remained concentrated near resonance regions and geometric interfaces.

To evaluate the contribution of individual architectural components, ablation experiments were performed by systematically removing or replacing major physics-aware modules (Fig. 4d). Removing wavelength-related phase conditioning, transformer bottleneck modelling or mixed-boundary convolution resulted in substantial increases in prediction error. Performance degradation was also observed when replacing GroupNorm, removing FiLM modulation or excluding geometry-aware distance fields. These results confirm the contribution of phase conditioning, mixed-boundary convolution and transformer modelling to field prediction accuracy.

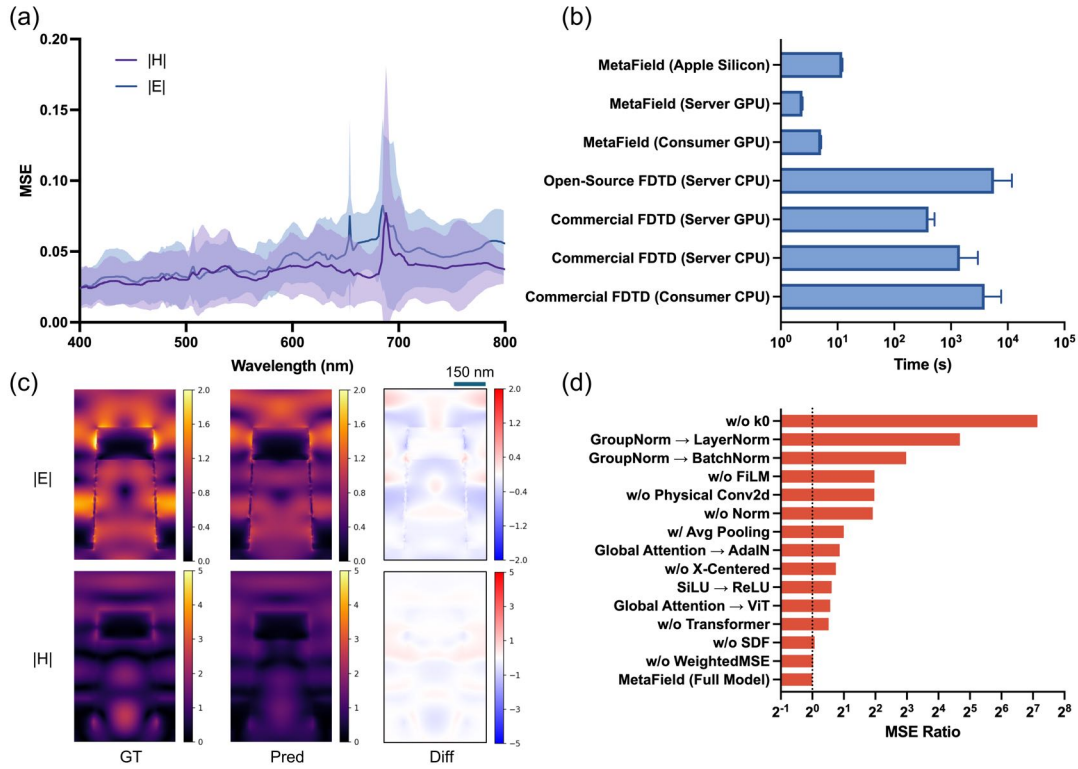


Figure 4 High-fidelity electromagnetic field prediction and computational acceleration. (a) Wavelength-dependent prediction error of the dominant electromagnetic field components. Solid lines indicate mean MSE and shaded regions indicate standard deviation across the validation dataset. (b) Inference-time comparison between MetaField and full-wave electromagnetic simulation frameworks on different hardware platforms. (c) Representative comparisons between ground-truth full-wave fields, MetaField predictions and residual field distributions. (d) Ablation analysis of the MetaField framework using normalized MSE ratios.

MetaField-guided inverse design

To evaluate the applicability of MetaField for large-scale nanophotonic optimization, the trained surrogate was integrated with a fabrication-aware genetic algorithm (GA) framework for field-guided inverse design (Fig. 5a). The workflow evaluated candidate structures using field-based fitness functions (Methods). Within the GA framework, fabrication-aware encoded structures were treated as optimization genomes containing geometry, material-stack and excitation-frequency parameters. For each candidate, MetaField predicted the corresponding electromagnetic field distribution, and the fitness value was computed from the top-5% near-field enhancement intensity within analyte-accessible regions. The evolutionary workflow

209 included elitism, selection, crossover and mutation operations while maintaining
 210 fabrication-aware structural constraints throughout the search process.

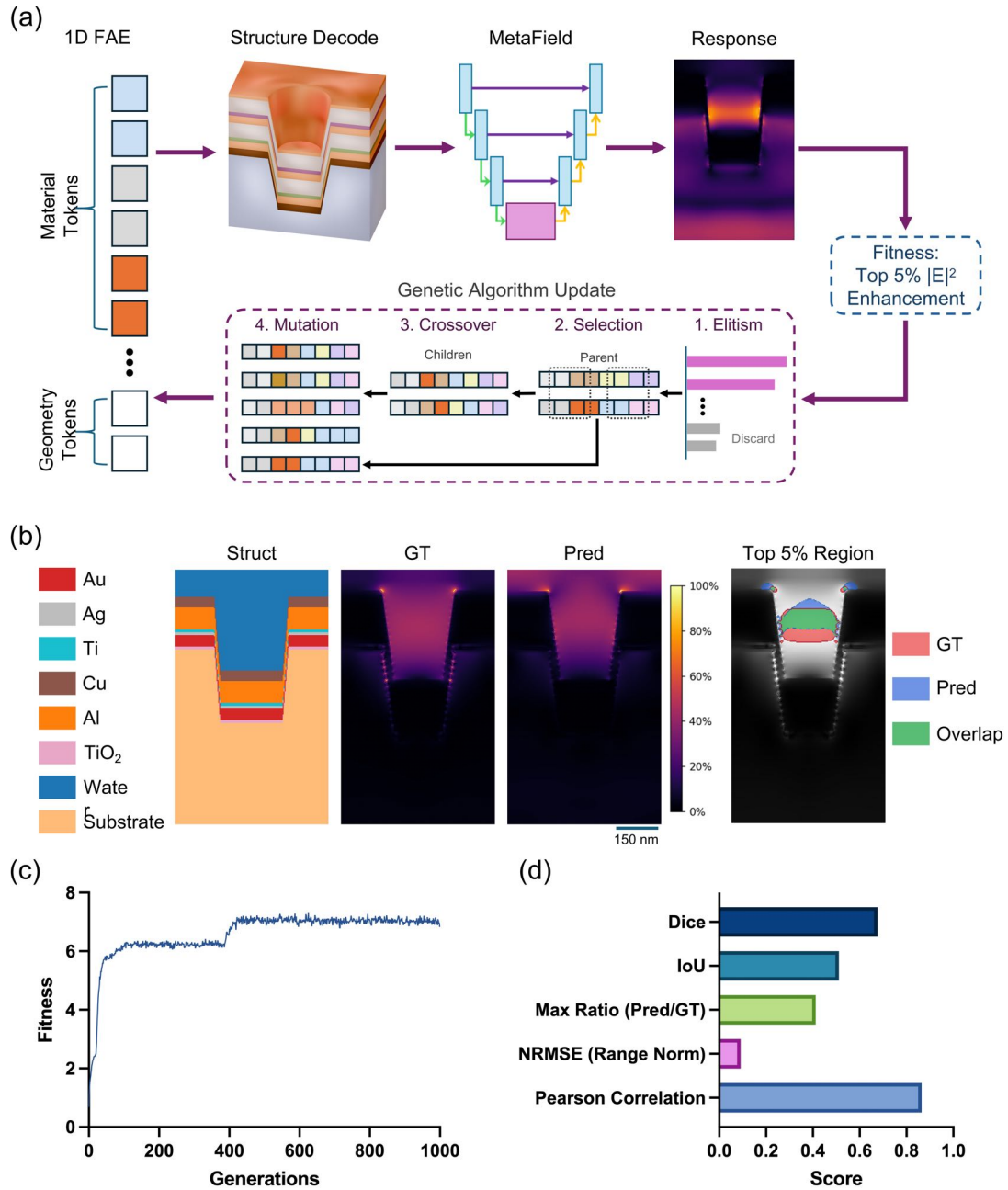


Figure 5 MetaField-guided inverse design. (a) Surrogate-assisted genetic-algorithm workflow for fabrication-aware inverse design. (b) Representative optimized structure together with full-wave ground-truth fields, MetaField predictions and hotspot-overlap comparison. (c) Genetic-algorithm convergence during large-scale surrogate-assisted optimization. (d) Quantitative comparison between MetaField predictions and full-wave FDTD results using field-correlation and hotspot-overlap metrics.

211 Figure 5b shows a representative optimized structure together with the
 212 corresponding full-wave FDTD field distribution, MetaField prediction and hotspot-

overlap analysis. MetaField preserved hotspot morphology with high Pearson correlation and stable overlap scores (Fig. 5d), indicating consistent reconstruction of dominant near-field regions despite local amplitude deviations.

The optimization convergence is shown in Fig. 5c. The surrogate evaluated 1,024,000 candidates in ~ 6 h, versus ~ 150 s per FDTD structure. Brute-force optimization would require $\sim 1.5 \times 10^8$ s, yielding an effective acceleration of $\sim 7 \times 10^3$.

SERS-oriented field analysis

To further evaluate the applicability of MetaField for nanophotonic sensing analysis, we compared MetaField-predicted near-field responses with experimentally measured SERS trends using nanopillar-based MetaSPR structures. Figure 6a shows the MetaField-predicted electromagnetic field distributions for two representative structures with different heights, denoted as Low and High. The High structure exhibited stronger hotspot confinement near the metal–dielectric interface than the Low structure, with correspondingly larger top-5% near-field enhancement intensity. The corresponding fabricated structures were subsequently characterized using SEM imaging (Fig. 6b).

Experimental SERS measurements were performed using $0.1 \mu\text{M}$ Rhodamine 6G (R6G) aqueous solution. As shown in Fig. 6c, the High structure consistently produced stronger Raman responses across Raman peaks. Quantitative comparison of Raman peak areas is summarized in Fig. 6d. The experimentally measured Raman enhancement trend followed the same relative ordering as the MetaField Prediction.

Rather than directly predicting Raman spectra, MetaField was used here as a field-oriented electromagnetic analysis framework. These results support field-guided comparative analysis and structure screening using MetaField.

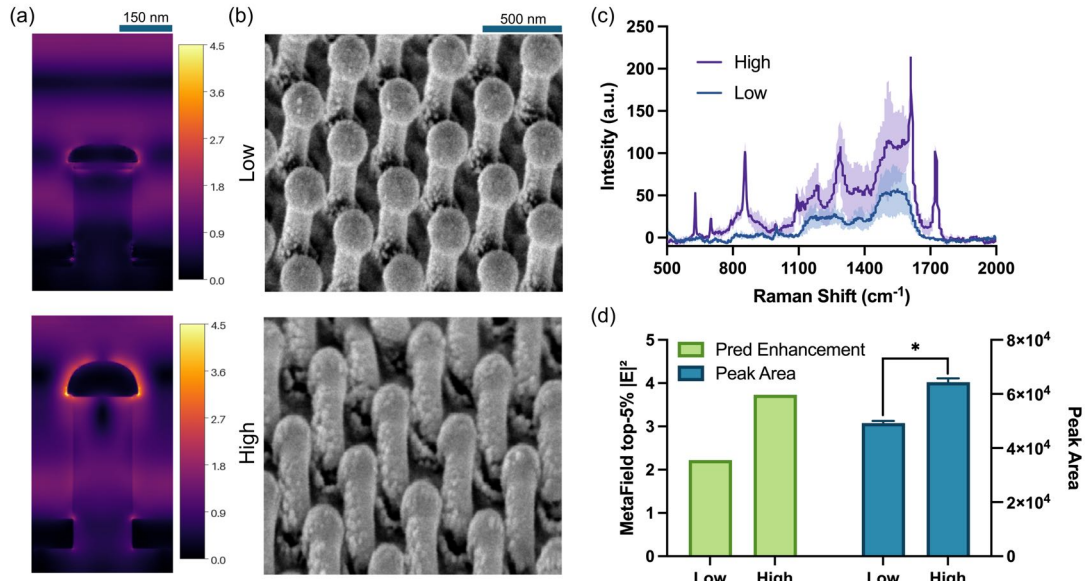


Figure 6 SERS-oriented field analysis and experimental comparison. (a) SEM characterization of Low- and High-height nanopillar MetaSPR structures. (b) MetaField-predicted near-field enhancement distributions for the corresponding structures. (c) Experimental SERS spectra measured using 0.1 μM Rhodamine 6G (R6G). Shaded regions indicate standard deviation across repeated measurements. (d) Comparison between MetaField-predicted field enhancement and experimentally measured Raman peak-area responses.

238 Conclusion

239 We introduced MetaField, a physics-aware neural surrogate that learns spatially
 240 resolved electromagnetic field distributions directly from fabrication-aware periodic
 241 MetaSPR structures, shifting metasurface machine learning from scalar-response
 242 prediction to field-resolved learning. MetaField combines a fabrication-aware
 243 simulation database (MetaSPR-SimDB), a neural architecture with mixed-boundary
 244 convolutions, conditional modulation and transformer bottleneck, and a field-guided
 245 workflow linking accelerated prediction to inverse design and experimental sensing
 246 analysis. Over the visible range, MetaField reproduces full-wave field distributions
 247 with orders-of-magnitude speed-up relative to FDTD, enables large-scale inverse
 248 design, and provides near-field indicators consistent with measured SERS trends. This
 249 framework is intended as a field-resolved surrogate for accelerated design exploration
 250 and rapid electromagnetic screening, complementing rather than replacing full-wave

solvers. Future work may extend MetaField to fully three-dimensional modelling, broadband multi-physics coupling and closed-loop experimental optimization.

Methods

FDTD simulation and data processing

Full-wave electromagnetic simulations were performed using the finite-difference time-domain solver in Ansys Lumerical FDTD Solutions. Periodic MetaSPR nanophotonic sensor structures were simulated under a two-dimensional x-z TM-like illumination setting over a wavelength range of approximately 400–800 nm. For each structure, complex refractive-index distributions and electromagnetic field components were recorded on geometry-resolved spatial grids. The dominant field components E_x , E_z and H_y were extracted together with the corresponding refractive-index distributions, spatial coordinates and wavelength information. Simulation outputs were exported as MATLAB mat files for subsequent data processing.

The exported mat file simulation files were converted into unified HDF5 datasets for physics-aware tensor construction. For each sample, complex refractive indices were transformed into permittivity channels according to:

$$\epsilon_r = n^2 - k^2, \quad \epsilon_i = 2nk$$

Geometry-aligned spatial padding was then applied to all samples, where horizontal boundaries used circular extension to preserve periodic continuity and vertical boundaries used replicate extension to maintain physically consistent field continuation along the propagation direction. Spatial coordinates were linearly extended to match the padded domains.

The processed HDF5 samples were grouped by spatial size and stored as MetaSPR-SimDB. Complex permittivity channels, spatial phase coordinates, geometry-aware normalized distance fields and global conditioning vectors (wavenumber, period-dependent phase, source-position phase) were assembled into physics-aware tensors as described in the main text. During training, the database was further converted to LMDB format for efficient distributed loading. The magnetic field

component H_y was additionally scaled by the free-space impedance $Z_0 = 376.73 \, \Omega$ to maintain numerical consistency with the electric-field channels.

MetaField architecture and training

MetaField used 96 base channels, eight attention heads (max relative-position distance 48) and eight transformer layers at the bottleneck. Training employed two NVIDIA A100 GPUs with distributed data parallelism and mixed precision; per-GPU batch size 36, four-step gradient accumulation (effective batch 288). The model was trained for 2000 epochs with AdamW (max lr = 6×10^{-4} , weight decay 10^{-4}) and a OneCycle schedule. The loss combined a masked, field-weighted MSE (weight = $1 + 5|E|$) and a gradient-consistency term (Sobel operators, weight 10^{-4}). Validation used masked MSE without field weighting. Metrics include range-normalized RMSE, Pearson correlation, and IoU/Dice for binarized hotspots. Inference throughput was benchmarked using synthetic input tensors of the same padded spatial size as the simulation data, with batch size increased exponentially until an out-of-memory or framework limit was reached. Five warm-up and 20 timed iterations per batch size were executed with device synchronization; throughput was reported as structures per second (excluding file I/O).

Genetic algorithm

MetaField was used as the surrogate evaluator in a genetic algorithm for inverse design. Each genome encoded excitation frequency, geometry and multilayer material sequence:

$$g = [f, R_{\text{top}}, R_{\text{bot}}, H, P, m_1, \dots, m_{50}]$$

The frequency search range was 374.74–749.48 THz, corresponding to approximately 800–400 nm. Geometry bounds were $100 \, \text{nm} \leq R_{\text{top}} \leq 180 \, \text{nm}$, $100 \, \text{nm} \leq R_{\text{bot}} \leq 150 \, \text{nm}$, $-500 \, \text{nm} \leq H \leq 500 \, \text{nm}$ and $300 \, \text{nm} \leq P \leq 600 \, \text{nm}$. A minimum lateral gap of 12 nm was enforced. The material genome contained 70 slots with a minimum block length of three cells and a maximum of five material transitions. The

first active material layer was constrained to an allowed adhesion-material set (Ti, TiO₂, Nb₂O₅). A repair function was applied after initialization, crossover and mutation to enforce geometric bounds, radius ordering, minimum gap, material-block length and transition constraints.

The GA used elitism, tournament selection, crossover and mutation. Unless otherwise specified, the population size was 1024, the number of generations was 1000. During crossover, each of the first five continuous genes was independently inherited from either parent, or a contiguous segment of the material genome was exchanged. During geometry mutation, one of the five continuous genes was perturbed by Gaussian noise with a standard deviation equal to 10% of its allowed range. During material mutation, a random block of material slots was replaced by a randomly selected allowed material.

For each candidate, MetaField predicted the electromagnetic field distribution, and the fitness was defined as the mean of the top 5% $|E|^2$ values within the fluid/analyte-accessible mask:

$$F = \text{mean}(\text{top}_{5\%}(|E|^2))$$

Experimental methods

Nanopillar-based MetaSPR sensor chips were fabricated via nanoimprint lithography from replicated nanocup master wafers. Two structural heights, denoted Low (~400 nm) and High (~650 nm), were prepared. For Low structures, a metallic coating of 10 nm Ti / 90 nm Au was deposited by physical vapor deposition; for High structures, 25 nm Nb₂O₅ / 50 nm Au was deposited.

Structural characterization was performed using a Quanta 3D FEG Dual Beam focused-ion-beam/scanning-electron-microscopy system. Images were used to reconstruct geometry-aware meshes for electromagnetic simulations and MetaField-based field analysis.

Surface-enhanced Raman spectroscopy measurements were conducted using a microscope-based confocal Raman spectroscopy system. Rhodamine 6G (R6G)

aqueous solution with a concentration of 0.1 μ M was used as the analyte. All samples were measured under identical optical acquisition conditions to enable comparative analysis between experimentally observed Raman enhancement trends and MetaField-predicted near-field responses.

Author Contributions

Mingqian Chen: Conceptualization, Methodology, Software, Formal Analysis, Validation, Visualization, Writing – Original Draft Preparation. Feiyang Wu: Data Curation, Investigation, Validation, Formal Analysis. Yuhong Li: Investigation, Resources, Validation. Shiyu Zhang: Visualization, Data Curation. Liping Huang: Supervision, Funding Acquisition, Writing – Review & Editing. Gang Logan Liu: Supervision, Resources, Funding Acquisition, Writing – Review & Editing. Wenjun Hu: Conceptualization, Supervision, Project Administration, Funding Acquisition, Writing – Review & Editing. Mingqian Chen, Feiyang Wu and Yuhong Li contribute equally to this paper.

Data Availability

MetaSPR-SimDB and associated preprocessing scripts are publicly available at Science Data Bank: <https://doi.org/10.57760/sciencedb.37386>. The MetaField source codes are publicly available at: <https://github.com/MrChen156/MetaField>. Trained model checkpoints will be publicly released upon acceptance.

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